

Corolario del orden de las normas $\mathcal{L}^p$

Corolario 1. Sean $q, p \in [1, \infty)$ con $p \leq q$ y $X \in \mathcal{L}^p(\mathbb{P})$

$$\implies X \in \mathcal{L}^q(\mathbb{P}) \wedge \|X\|_p \leq \|X\|_q.$$

Demostración: El caso $p = q$ es trivial. Supongamos $p < q$ y definimos

$$\varphi(t) := \begin{cases} t^{q/p} & \text{si } t \geq 0 \\ 0 & \text{si } t < 0 \end{cases} \xrightarrow{q > p} \varphi \text{ es } \mathcal{C}^1 \text{ y convexa.}$$

Entonces, por la desigualdad de Jensen, tenemos que

$$\varphi(\mathbb{E}[|X|^p]) \leq \mathbb{E}[\varphi(|X|^p)] = \mathbb{E}[|X|^q] = \|X\|_q^q$$

Ahora bien, como $\varphi(\mathbb{E}[|X|^p]) = (\mathbb{E}[|X|^p])^{q/p} = \|X\|_p^q$, se tiene que $\|X\|_p \leq \|X\|_q$. ■

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